



TIIPE Digital Platform: Unified Web Portal & Mobile App

PRODUCT REQUIREMENTS DOCUMENT

Transforming knowledge into impact by expanding access to education, lifelong learning, workforce development, public health education, civic engagement, and community-based research.

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1. Context and Why We Are Building This

Low-income and underserved populations in New Rochelle, NY and globally encounter persistent structural barriers to economic mobility, academic success and healthcare. These barriers manifest as a lack of tutoring resources for students, limited workforce readiness training, and a critical shortage of culturally relevant health education. The Impact Institute for Public Education (TIPE), Inc. addresses these challenges by expanding access to learning, mentorship, research-driven tools, and public interest outreach.

Currently, TIPE's digital presence consists of a basic website (<https://impactinstitute.global.org/>). While this site provides introductory information, it does not serve as an active workspace or resource delivery system. To maximize impact, TIPE requires a comprehensive, unified digital ecosystem. This platform must serve two key purposes: first, a redesigned Web Portal that establishes public trust, facilitates governance transparency, accepts donations and provides research; second, a brand-new Mobile Application (built with Flutter) that connects learners and mentors, delivers health education campaigns and tracks progress.

The Single Problem We Are Solving:

Underserved learners and community advocates have no integrated, low-friction digital platform to access educational mentorship, public health literacy and civic participation resources, while donors lack a direct, transparent channel to fund and see the immediate local impact of these specific programs.

2. Mission and Vision

Our Mission

Our mission is to promote equitable access to education and lifelong learning by offering tutoring, mentorship, workforce development, health education, civic engagement, research, and community-based programs with a focus on underserved and low-income populations.

Our Vision

A world where equitable access to education, knowledge, opportunity and civic participation empowers individuals and communities to thrive and create lasting social impact.

Brand Personality

Three defining qualities must govern the user experience of all TIPE digital products:

Value	What It Means in Practice
Empowerment	Every UI flow must lead to an actionable,

	positive outcome. Learners should feel encouraged by clear progress markers, volunteers should feel valued and donors should see the immediate impact of their contributions.
Inclusiveness	The design must accommodate users of varying digital literacy, physical accessibility (WCAG 2.1 AA) and bandwidth constraints. Layouts must be simple, readable and lightning-fast.
Transparency	Governance, nonprofit credentials, and donation allocations must be prominently highlighted. Users should be able to access financial reporting and policy briefs in less than two clicks.

3. Who We Are Building For

User Class	Core Profile	Primary Digital Need
The Learner	Underserved youth, low-income students and adults seeking digital literacy or career training.	A free mobile app that runs on low-bandwidth, supports offline reading, lists local courses/tutoring and issues certificates.
The Mentor / Volunteer	Professionals, teachers and civic leaders looking to donate their expertise.	Low-friction signup, profile verification, interactive schedule builder to match with students and mentorship resources.
The Donor	Philanthropic individuals, corporate partners and institutional grant representatives.	A secure web portal with one-click or monthly recurring giving options, tax receipt generation and real-time funding dashboards.
Community Partners	Nonprofits, schools, health agencies and local businesses collaborating with TIIPE.	A central portal to co-manage programs, upload educational materials and access community-based research.

4. What We Are Building

We are building a unified digital platform consisting of the Web Portal and the TIIPE Mobile App. To ensure high engineering velocity, features are segmented into three distinct phases.

PHASE 1 — BUILD NOW (Months 1 to 3)

Focus: Develop the web portal, integrate donations, publicize governance and set up basic user auth.

- **Web Develop:** Responsive portal with updated branding, hero section, CTA buttons and responsive pages.
- **Donation Engine:** Secure Stripe/PayPal donation widget with one-time, monthly recurring, and program-targeted options.
- **Governance Center:** Bylaws, Conflict of Interest policy, 501(c)(3) status and Board Bios accessible to download.
- **Auth and Profiles:** Onboarding and profiles for Learners, Mentors, and Administrators.
- **Tutoring MVP:** A schedule-matching system where Learners can request academic tutoring sessions with verified Mentors.
- **Media Hub:** Embedding AdieTalk Radio, podcasts and upcoming webinars directly into the site.

PHASE 2 — BUILD LATER (Months 4 to 6)

Focus: Launch the Flutter mobile app, introduce mobile-specific modules and automate volunteer matching.

- **TIPE Mobile App:** Cross-platform iOS and Android app with push notifications and basic offline database storage.
- **In-App Chat:** Real-time messaging with attachment sharing between matched Mentors and Learners.
- **Micro-Learning Modules:** Interactive, self-paced learning pathways for digital literacy and workforce development.
- **Public Health Portal:** A section in the app for health trackers, maternal health literacy materials and disease prevention updates.
- **Matching Algorithm:** Matching volunteers with students based on availability, subject matter expertise and location.

PHASE 3 — SCALE (Months 7 to 12)

Focus: AI integrations, open-access research portal, global outreach and developer APIs.

- **AI Learning Copilot:** AI-driven chatbot helping students find specific training modules and answering basic educational queries.
- **Research Portal:** Searchable academic repository for community-based research, policy briefs and educational journals.
- **Developer APIs:** A public API enabling partner schools to access TIPE's digital curriculum and scheduling software.

5. Design Requirements

Design Principle	Implementation Rule
Inclusive Typography & Palette	Deep Navy Blue (#1E3A8A) for trust, Vibrant Amber (#D97706) for call-to-actions, Forest Green (#059669) for health pages. Fonts: Outfit (headings), Inter (body, minimum 16px).
Responsive & Mobile-First	Optimize for 375px viewport first. Ensure fast performance on low-end smartphones. Limit layout elements to single columns on mobile devices.
Clear Accessibility (WCAG 2.1 AA)	Ensure color contrast ratio of at least 4.5:1 for all text. Provide clear ARIA landmarks, alt text on images, and full keyboard navigability.
Speed and Perception	Ensure loading times below 2 seconds. Use skeleton states, immediate CSS transitions, and local caching of assets on the mobile application.

6. Project Timeline

Phase / Timeframe	Key Deliverables	Engineering Owner
Weeks 1 - 2	Define requirements, set up database schemas (PostgreSQL), initialize project folders, Paystack/Stripe sandbox configurations.	Lead Architect + DB Dev
Weeks 3 - 5	UI/UX wireframes for the portal and mobile app screens. Core landing page layout coding.	UX Lead + Frontend Dev
Weeks 6 - 8	Stripe payment integration, Donation portal deployment, User login and Profile creation systems.	Backend Dev + Frontend Dev
Weeks 9 - 10	Governance center upload/download systems, AdieTalk Radio and webinars embedding, MVP tutoring schedule.	Fullstack Engineer

Weeks 11 - 12	System QA, load testing on the checkout system, user acceptance testing with 10 community pilot users, soft launch.	QA Lead + DevOps
Months 4 - 6	Build and launch the Flutter Mobile App (iOS/Android), matching algorithm, in-app chat, health literacy portal.	Mobile Dev + Backend Dev

7. Budget Estimate in USD

Expense Category	Description / Tool	Annual Cost (USD)
Cloud Infrastructure	AWS Hosting (EC2, RDS PostgreSQL, S3 for documents), Vercel for Frontend.	\$-
Developer Accounts	Apple Developer Program (\$99/yr), Google Play Console (\$25 one-time).	\$-
Third-Party Integrations	SendGrid (transactional email), Twilio (SMS notifications), SSL Certificates.	\$-
Development Team	Contract fees for 1 UX Designer, 1 Mobile/Flutter Dev, 1 Backend Dev (Phase 1-2).	\$-
Compliance & Security	Background check API (Checkr) for mentor validation (\$- per tutor, estimate 100 tutors), legal audit.	\$-
TOTAL ESTIMATE	Initial development, deployment and 1 year of active platform maintenance.	\$-

8. Donation, Funding, and Sustainability Model

As a 501(c)(3) charitable nonprofit organization, TIPE relies on philanthropy, public-interest partnerships and corporate matching. The digital platform is designed to build a sustainable funding flywheel through:

- **Targeted Program Donating:** Enabling donors to direct funds to specific initiatives (e.g., general funds, tutoring kits, public health brochures).
- **Monthly Giving Program:** Integrating a recurring donation checkout to secure monthly predictable operational funding.
- **Transparency & Impact Tracking:** Providing downloadable impact PDFs after donation, showing where the money goes.
- **Transaction Fees:** TIPE is eligible for discounted Stripe/PayPal processing rates (typically 2.2% + \$0.30 per transaction for registered 501(c)(3) organizations). The system will automatically pass transaction logs to the financial ledger.

9. How We Will Measure Success

Category	Primary Success Metric	12-Month Target
Education & Mentoring	Number of active learners onboarded and hours of verified tutoring completed.	2,000 learners; 5,000 hours of tutoring
Public Health Outreach	Downloads of health pamphlets and participation in health literacy webinars.	10,000 downloads; 1,500 webinar attendees
Civic Engagement	Monthly active listeners of AdieTalk Radio and policy briefs downloaded.	5,000 listeners; 500 policy briefs downloaded
Financial Sustainability	Gross Donation Value (GDV) and number of monthly recurring donors.	\$100,000 GDV; 300 recurring donors

10. Objectives and Key Results (OKRs)

Objective	Key Results	Owner
O1: Deliver a world-class digital learning experience for underserved youth.	KR 1.1: Onboard 500 verified mentors by Q2. KR 1.2: Match 1,000 students to tutors via mobile app. KR 1.3: Maintain user satisfaction rating above 90%.	Education Director
O2: Broaden public health and civic awareness through digital media.	KR 2.1: Transition AdieTalk Radio to a fully integrated digital podcast library. KR 2.2: Publish 6 community-based research policy briefs.	Outreach & Media Lead

	KR 2.3: Reach 10,000 unique downloads of resources.	
O3: Secure stable funding for the organization's growth.	KR 3.1: Launch Stripe donation widget with zero transaction loss in pilot. KR 3.2: Onboard 200 recurring donors. KR 3.3: Achieve \$50,000 in general public donations.	Treasurer / Executive Dir

11. User Personas and Journeys

To guide development, we outline three key personas representing our target audience:

Persona 1: Marcus, The Learner

Profile: 17-year-old student residing in low-income housing in New Rochelle. He has limited access to computers and relies entirely on his smartphone with a prepaid data plan. Marcus struggles with math and wants to enter a coding bootcamp next year.

User Journey:

1. Marcus downloads the TIPE mobile app on his phone.
2. He signs up for a Learner account and sets his focus area to Mathematics and Digital Technology.
3. He browses available tutors, finds Sarah, and requests a 1-hour weekend slot.
4. He receives an automated SMS confirmation with a link to join the chat.
5. After the session, Marcus marks the tutoring session as complete and downloads the coding worksheet to study offline.

Persona 2: Saratu, The Mentor / Volunteer

Profile: 29-year-old software engineer living in New York. She wants to volunteer 2 hours a week to mentor underrepresented youth in technology but struggles to find structured programs that fit her busy work schedule.

User Journey:

1. Sarah visits the TIPE Web Portal, navigates to the 'Support Us' page, and selects 'Become a Mentor'.
2. She fills out the volunteer application and uploads her ID for background screening.
3. Once verified, she logs into her Mentor Dashboard and inputs her weekly availability.
4. She receives an email notification when Marcus requests a slot. She clicks 'Accept'.
5. The portal generates a video link. After the session, she leaves a brief progress note for Marcus.

Persona 3: Helen, The Donor

Profile: 45-year-old local business owner committed to community development. She values financial transparency and wants to know exactly how her donations are spent before committing to a monthly contribution.

User Journey:

1. Helen visits impactinstitute.org to research the organization.
2. She reads the Governance section, downloads the Bylaws and recent Form 990 to verify the nonprofit's compliance.
3. Satisfied, she navigates to the 'Donate' portal, selects 'Monthly Giving', and sets up a \$100 recurring payment.
4. She receives an automated tax receipt email and a link to the real-time funding impact dashboard.

12. Competitive Analysis

Platform	Core Strengths	Core Weaknesses	TIIE Competitive Edge
VolunteerMatch	Massive reach; connects volunteers to thousands of opportunities.	Very generic; no direct learning management or mentoring platform built-in.	TIIE offers a custom, closed-loop mentorship matching engine and chat tools built directly into the app.
Khan Academy	World-class self-paced academic tutoring and video library.	Lacks direct human connection, custom mentor scheduling, or local community research.	TIIE integrates human-to-human mentoring with local community engagement and specific workforce paths.
Charity Navigator	High-trust registry showing nonprofit transparency scores.	Only lists directory data; does not offer donation checkouts or direct program management.	TIIE provides a public, self-serve Governance center alongside direct program impact dashboards.
TIIE Platform	Combined tutoring scheduling, volunteer portal, donation gateway, media library (AdieTalk), and public health tracker in one.	Currently starting with a small local pilot group; requires active marketing to build initial base.	Provides a hyper-local, high-trust ecosystem connecting local New Rochelle needs to global resources.

13. Information Architecture and Screen Descriptions

13.1 Platform Structure (Web and Mobile)

Section	Contents and Function
Marketing Homepage (Web)	Hero Section with CTA buttons, About Us teaser, core program summaries, public benefit statement, and footer links.
Programs Portal (Web & App)	Dedicated landing pages for the 4 core initiatives. Contains focus areas, volunteer details, and links to register.
Governance Center (Web)	Public-facing dashboard showing Board of Directors, executive profiles, ethical principles, conflict of interest recusal standards, and bylaws PDFs.
Support Us / Donate (Web & App)	Secure credit card/PayPal checkout, monthly donation setup, general fund allocation options, and tax receipt generator.
Member Dashboard (App & Web)	Post-login view. Learners see upcoming tutoring, resources, and chats. Mentors see schedules, student requests, and session logs.
Media Hub (Web & App)	Embeds AdieTalk Radio player, podcast directory, registration links for webinars, and research publications archive.

13.2 Key Screen Descriptions for Design

- **Web Homepage:** Redesigned Web Homepage: A modern, high-conversion landing page. The top header houses a navigation bar (Home, About Us, Programs, Governance, Support Us, Contact, Donate). A prominent Hero section features the Headline, Subheadline, and three colored CTA buttons. A counter displaying key impact metrics (e.g. '500+ Students Mentored') adds credibility, followed by the 'Who We Are' section and a footer.
- **Governance Screen:** Governance & Downloads Screen: A dedicated transparency page. It contains the Governance Overview and lists the Board of Directors with their bios. It features a principles section detailing Ethics, Accountability, Transparency, and Public Benefit, followed by a "Document Downloads" block containing Bylaws, Conflict of Interest policy, and 501(c)(3) tax status documents.
- **Donation Portal:** Donation Checkout Screen: A secure, itemized form. The donor selects their giving type (one-time or monthly) and inputs an amount. They can allocate their donation to specific funds (General, Education, Health, Civic). The payment section uses

secure elements (Stripe/PayPal). Upon transaction success, the donor is shown a thank-you page with an option to download a tax-deductible receipt.

- **Mobile Dashboard:** Mobile App Workspace: The primary dashboard for Learners and Mentors. It features a clean top navigation with quick links to “Upcoming Sessions”, “Resources”, and “Chats”. The center contains card components for course pathways, a scheduling wizard, and a bottom navigation bar for quick access.

14. Data Model

The PostgreSQL schema for the TIPE platform manages user authentication, profile data, program enrollments, scheduling, donations and governance documents.

Database Entity	Key Fields and Relationships
USER	ID (PK), email, password_hash, role (Learner, Mentor, Admin), created_at, last_login.
PROFILE	ID (PK), user_id (FK), first_name, last_name, phone, bio, status (Active/Pending), background_check_status.
PROGRAM	ID (PK), title, description, category (Education, Health, Civic, Partnership), status, cover_image_url.
SESSION	ID (PK), learner_id (FK), mentor_id (FK), start_time, end_time, status (Requested, Scheduled, Completed, Cancelled), meeting_link, feedback_notes.
DONATION	ID (PK), donor_email, amount, frequency (One-time/Monthly), fund_allocation (General, Education, Health, Civic), transaction_id, status (Success/Failed), invoice_url.

15. Donation and Funds Allocation Flow

All transactions are processed through Stripe or PayPal. When a donation is made, the funds are categorized in the database to ensure financial tracking and transparency.

Stage	Action and System Behavior
1. Input	Donor selects donation type, amount, and specific program allocation (e.g., \$100 allocated to 'Education Fund').

2. Gateway	Stripe securely processes the payment. A webhook notifies the TIIPE backend of the successful charge.
3. Logging	The backend creates a record in the DONATION table, generating a PDF invoice containing TIIPE's 501(c)(3) tax details.
4. Ledger	Funds are allocated to the corresponding sub-account in the accounting database (e.g. 100% of donation minus Stripe transaction fee is credited to the Education Fund ledger).
5. Payout & Report	The finance team updates the public impact metrics monthly, showing how much money was allocated, spent and the outcomes achieved.

Worked Example: A donor gives \$100 online.

Gross donation: \$100.00

Stripe nonprofit transaction fee (~2.2% + \$0.30): \$2.50

Net donation transferred to TIIPE bank: \$97.50

Amount allocated directly to Education Program: \$97.50

The donor's impact report will reflect that \$97.50 funded textbook bundles and online tutoring software licenses.

16. Compliance and Data Privacy

- **COPPA Compliance (US):** Because the platform provides tutoring for high school students, the site and app must strictly comply with the Children's Online Privacy Protection Act (COPPA). No personal data of children under 13 may be collected without verifiable parental consent.
- **GDPR / NDPR Compliance:** To support international learners and donors, the platform will implement GDPR and NDPR protocols. Users can request data erasure, opt-out of tracking, and view our standard Data Protection Agreement (DPA).
- **PCI DSS (Payments):** All payment checkouts are fully outsourced to Stripe/PayPal. The platform never stores raw credit card numbers, operating under PCI-DSS SAQ-A guidelines.
- **Mentor Background Checks:** All mentors must clear a background check (integrated via Checkr API) before they are permitted to schedule sessions with students. Administrative reviews are logged for audit compliance.

17. Analytics and Retention Strategy

We track platform engagement to optimize resource delivery and minimize user drop-off:

- **Student Retention:** Monitoring active learners, completed tutoring hours and certificate completion rates.
- **Volunteer Engagement:** Tracking volunteer signups, active schedules and average duration of mentorship cycles.
- **Donor Analytics:** Tracking recurring donor retention rates, average gift size and monthly active donor counts.
- **Retention Loops:** TIPE will implement email newsletters, volunteer recognition badges, and automated session follow-up emails containing student progress summaries.

18. Go-to-Market Plan

To build our user base in New Rochelle and globally, TIPE will implement a multi-channel outreach plan:

- **Local School Partnerships:** Partnering with New Rochelle high schools, public libraries, and community centers to distribute app flyers and register students.
- **Volunteer Recruitment:** Recruiting professional mentors through targeted LinkedIn outreach, corporate CSR partnerships, and college education departments.
- **Media and Content Strategy:** Syndicating Adietalk Radio, launching podcasts, hosting public webinars and publishing policy research papers on social media.

19. Full Feature Prioritisation Table

Feature Description	Category	MoSCoW State	Target Release
Responsive Web Portal Development & Branding	Core Web	Must Have	Phase 1
Stripe/PayPal Donation Checkout	Donation	Must Have	Phase 1
Governance Downloads & Board Bios	Governance	Must Have	Phase 1
User Auth, Profiles, and Tutoring Scheduler MVP	LMS	Must Have	Phase 1
TIPE Mobile App (iOS/Android) with in-app chat	Mobile App	Should Have	Phase 2
AI Learning Copilot /	AI Integration	Could Have	Phase 3

Chatbot			
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20. 18-Month Product Roadmap

Quarter	Core Theme	Key Milestones & Deliverables
Q1 - Q2 (Months 1 - 6)	Core Web development & Mobile MVP	Deploy developed Web Portal, Governance downloads and Donation widget. Build and launch the initial Flutter Mobile Application with chat and tutoring scheduling.
Q3 - Q4 (Months 7 - 12)	Engagement & Public Health Scaling	Integrate the public health portal and maternal health trackers on mobile. Automate mentor-learner matchmaking. Roll out the volunteer recognition system.
Q5 (Months 13 - 15)	Open Access & Partner Integration	Deploy the open-access Research Portal for policy briefs. Develop the public API for partner schools and local nonprofits.
Q6 (Months 16 - 18)	AI-Driven Personalization	Integrate the AI Learning Copilot to help students plan custom learning paths. Launch automated outreach to recurring donors with personalized impact reporting.

21. Decisions Still to Be Made

- **Video Tutoring Technology:** Evaluating Zoom API, Google Meet API, or open-source Jitsi SDK for hosting video tutoring sessions directly in the app.
- **LMS Architecture:** Choosing between a third-party LMS backend (like Moodle/Canvas) or developing a custom PostgreSQL course database.
- **Zero-Rated Data Partnerships:** Partnering with regional telecom providers to establish zero-rated data access for the TIPE mobile app in developing countries.

22. Risks and Mitigations

Risk Description	Likelihood	Impact	Mitigation Action
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COPPA compliance failure regarding minor profiles.	Low	High	Incorporate strict parental consent flows and age-gate verification during Learner signup. Audit the database quarterly.
Low volunteer mentor retention leading to long student waitlists.	Medium	Medium	Develop the volunteer badge/recognition program, coordinate monthly mentor meetups and automate scheduling notifications.
Stripe/PayPal gateway failure during high-traffic campaigns.	Low	High	Implement dual gateway support (both Stripe and PayPal) with fallback routing and automatic system alerts to administrators.
Offline data sync errors on mobile devices with poor connection.	High	Medium	Configure SQLCipher local database storage on mobile. Sync data in batches only when a stable internet connection is detected.

23. Document Authorization & Sign-off

By signing below, the stakeholders agree that this Product Requirements Document accurately reflects the functional requirements, design constraints and technical scope of the TIPE digital platform.

Stakeholder Name & Title	Signature	Date
President, TIPE Inc.		
Treasurer, Board of Directors		
Product & Technology Lead		

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